

LAB 3

Component Modelling & Architectural Pattern Selection

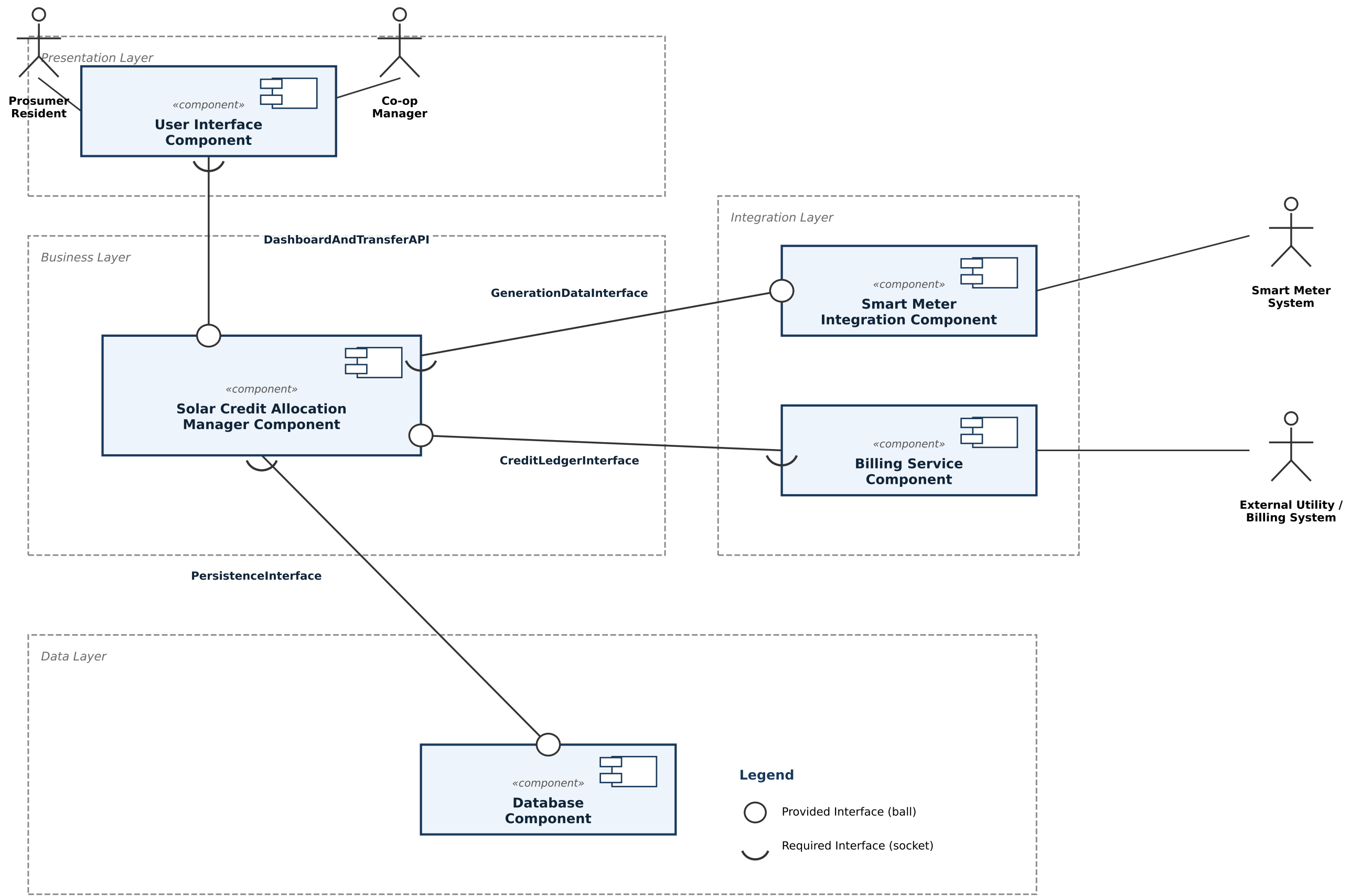
Community Solar Credit Allocation Manager
Problem Statement #63 | Sustainability & Green Tech

Name: C DINESH KUMAR

SRN: PES1UG24CS117

Community Solar Credit Allocation Manager — UML Component Diagram

Layered Architecture | Problem Statement #63 | PES1UG24CS117



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Architecture Selection

We chose Layered Architecture for the Community Solar Credit Allocation Manager system. The system is organised into a Presentation Layer (User Interface Component), a Business Layer (Solar Credit Allocation Manager Component), an Integration Layer (Smart Meter Integration Component and Billing Service Component), and a Data Layer (Database Component).

Two Scenario-Related Reasons

- Clear separation of concerns for credit logic: Solar generation recording, credit-transfer validation, recipient-eligibility checks, and billing-offset calculation are all core, evolving business rules (FR001-FR005). Isolating this logic in a single Business Layer component keeps it independent of how prosumers view the data (UI) and how it is stored (Database), so allocation rules can change without touching the presentation or persistence code.
- Clean isolation of two very different external integrations: the system must ingest continuous hourly telemetry from up to 500 Smart Meters (NFR001) while also exchanging periodic billing statements with an External Utility/Billing System. A layered design routes both through dedicated Integration Layer components (Smart Meter Integration, Billing Service) with well-defined interfaces, so each integration can change, fail, or be replaced independently without destabilising the core allocation logic.

Security Advantage

Because every credit transfer, balance check, and recipient-eligibility verification must pass through the single Solar Credit Allocation Manager Component in the Business Layer, the system has one centralised enforcement point for access control and transaction rules (NFR002). Authentication and authorisation checks are not duplicated or scattered across the UI or Database layers, which reduces the risk of an inconsistent or bypassable permission check and keeps sensitive credit and billing data reachable only through validated business-layer calls.

Performance Benefit

Routing all smart-meter telemetry through a dedicated Smart Meter Integration Component lets incoming generation data be validated, batched, and queued at the Integration Layer boundary before it reaches the Business Layer. This keeps the high-frequency, high-volume meter feed from blocking prosumer-facing operations such as viewing credit balances or initiating a transfer, helping the system sustain the 99.9% uptime target for processing hourly data from 500 households (NFR001) while still responding promptly to interactive requests.